CLASS 8 · CBSE · GANITA PRAKASH PART 1 · SHORT NOTES

Chapter 3: A Story of Numbers

Topic-wise short notes with solved examples

How people learned to count and write numbers: tally marks, grouping, landmark numbers, the idea of a base, place value and zero, ending with the Hindu (Indian) number system used across the world today.

1 Counting and number systems

- To count a collection we match its objects one by one with a fixed, ordered sequence. This matching is a **one-to-one mapping**: no two objects are matched with the same item.
- The fixed sequence can be made of **objects** (sticks, pebbles), **names** (spoken words) or **written symbols**. Such a standard sequence is called a **number system**.
- The symbols of a written number system are called **numerals**.
- A good number system must never run out (numbers have no end) and must still be easy to use.

Method	Strength	Weakness
A stick for each object	Never runs out	Needs as many sticks as objects
Letters a to z as names	Easy to say	Stops at 26
Roman symbols	Short for small numbers	Needs new symbols for larger numbers

Example 1.1 A herder keeps one pebble in a bag for each sheep that leaves in the morning. In the evening she removes one pebble for each sheep that returns, and 3 pebbles are left. What does this tell her?

SOLUTION

Each pebble is matched with exactly one sheep.

3 sheep have not returned. She learns this without using any number names.

2 Early ways of counting

- **Body parts**: some communities count by pointing to fingers, wrist, elbow, shoulder and so on in a fixed order.
- **Tally marks**: one notch for each object. The Lebombo bone (about 44,000 years old, 29 notches) and the Ishango bone (20,000 to 35,000 years old) are the oldest known examples.
- **Counting in twos**: the Gumulgal people of Australia used *urapon* for 1 and *ukasar* for 2, and built other names from them: $3 = 2 + 1$ is ukasar-urapon, $4 = 2 + 2$ is ukasar-ukasar, $5 = 2 + 2 + 1$, $6 = 2 + 2 + 2$.
- The key idea is **grouping**: count in groups of a fixed size (commonly 2, 5, 10 or 20) and use the group to name larger numbers.

- Most people can tell the size of a collection at a glance only up to about 4. This may be why tally marks are bundled in fives.

Example 2.1 Write 9 in the Gumulgal style, counting in twos.

SOLUTION

$9 = 2 + 2 + 2 + 2 + 1$

ukasar-ukasar-ukasar-ukasar-urapon

3 Roman numerals

- The Roman system uses several group sizes at once. Numbers that have their own symbol are called **landmark numbers**.

Symbol	I	V	X	L	C	D	M
Value	1	5	10	50	100	500	1000

- To write a number, break it into landmark numbers, taking as many of the largest as possible first.
- A smaller symbol written **before** a larger one is subtracted: IV $= 4$, IX $= 9$, XL $= 40$, XC $= 90$, CD $= 400$, CM $= 900$.
- Weaknesses: there is no zero, larger numbers need new symbols, and multiplication and division are very hard.

Example 3.1 Write 1947 in Roman numerals.

SOLUTION

$1947 = 1000 + 900 + 40 + 7$

M, CM, XL, VII

MCMXLVII

Example 3.2 Add CCXXXII and CCCCXIII without changing them to Hindu numerals.

SOLUTION

Collect the symbols: six Cs, four Xs and five Is.

Five Cs make a D, and five Is make a V.

DCXLV (that is, $232 + 413 = 645$)

4 The idea of a base

- The Egyptians (about 3000 BCE) chose landmark numbers that are all powers of 10: $1, 10, 100, 1000, \ldots$, with a separate symbol for each. A number was written by repeating each symbol as often as needed.
- A system whose landmark numbers are $1, n, n^2, n^3, \ldots$ is a **base-n number system**. Base 10 is also called the **decimal** system.

- In a base-n system, n copies of one landmark number always make the next one. So no symbol is ever needed n or more times.
- **Why a base helps**: the product of two landmark numbers is again a landmark number ($n^a \times n^b = n^{a+b}$). This makes addition and multiplication systematic. Roman landmark numbers do not have this property.
- Drawback of the Egyptian system: every higher power of 10 needs a brand-new symbol.

Example 4.1 What are the first five landmark numbers of a base-5 system? Break 143 into these landmark numbers.

SOLUTION

$5^0 = 1, 5^1 = 5, 5^2 = 25, 5^3 = 125, 5^4 = 625$

$143 = 125 + 5 + 5 + 5 + 1 + 1 + 1$

One 125, no 25s, three 5s and three 1s

Example 4.2 In the Egyptian system, how many symbols in all are needed to write 2304?

SOLUTION

$2304 = 2 \times 1000 + 3 \times 100 + 0 \times 10 + 4 \times 1$

$2 + 3 + 0 + 4 = 9$ **symbols**

- The **abacus** used in Europe from about the 11th century was a board with one line for each power of 10. Counters on the lines showed how many of each power were present.

5 Place value

- In a **place value system** (positional system) the **position** of a symbol tells which landmark number it counts. No separate symbols for the landmark numbers are needed, so a few symbols can write every number.

System	Base	Notes
Mesopotamian (Babylonian)	60	Symbols for 1 and 10 made the numbers 1 to 59; blank space for an empty place, later a placeholder symbol
Mayan	20, with places 1, 20, 360, ...	Dot for 1, bar for 5, a shell symbol for zero; written top to bottom
Chinese rod numerals	10	Rod patterns for 1 to 9, alternating upright and sideways forms; blank for an empty place
Hindu (Indian)	10	Ten digits 0 to 9, with 0 a full digit and a number

- The base-60 system survives in our units of time: 1 hour $= 60$ minutes and 1 minute $= 60$ seconds.
- Without a symbol for an empty place, a numeral can be read in more than one way. A **placeholder (zero)** removes this confusion.

Example 5.1 Break 7530 into Mesopotamian places.

SOLUTION

$7530 = 2 \times 3600 + 5 \times 60 + 30$

The three places hold 2, 5 and 30

Example 5.2 Break 500 into Mayan places (1, 20, 360).

SOLUTION

$500 = 1 \times 360 + 7 \times 20 + 0 \times 1$

The places hold 1, 7 and 0, written from top to bottom

6 The Hindu number system

- It is a base-10 place value system with ten digits $0, 1, 2, \ldots, 9$. For example $375 = 3 \times 10^2 + 7 \times 10 + 5 \times 1$.
- It began in India about 2000 years ago. The Bakhshali manuscript (about the 3rd century CE) writes zero as a dot. Āryabhaṭa (499 CE) computed extensively with the system.
- Brahmagupta (628 CE) gave rules for arithmetic with **zero as a number** and with negative numbers.
- It reached the Arab world by about 800 CE through Al-Khwārizmī and Al-Kindi, and Europe by about 1100 CE. Fibonacci argued for it around 1200, and it was in general use in Europe by the 17th century.
- Europeans learned the numerals from Arab scholars and called them "Arabic numerals". Arab scholars themselves called them Hindu numerals. Today they are called Hindu, Indian or Hindu-Arabic numerals.
- Because zero is a digit and each place holds exactly one digit, every number can be written in exactly one clear way, and calculation is easy.

Stage	Idea
1	Count in groups of one size
2	Group using landmark numbers
3	Make the landmark numbers the powers of one number: a base
4	Let position show the landmark number: place value
5	Use 0 as a digit and as a number

7 Writing numbers in other bases

- The same place value idea works for any base n, using the digits 0 to $n - 1$.
- To convert a number to base n: take out the largest power of n as many times as possible, then the next power, and so on. The counts are the digits.
- To convert back: multiply each digit by its power of n and add.

Example 7.1 Write 25 in base 8, base 5 and base 2.

SOLUTION

Base 8: $25 = 3 \times 8 + 1$, giving 31.

Base 5: $25 = 1 \times 25 + 0 \times 5 + 0$, giving 100.

Base 2: $25 = 16 + 8 + 1 = 1 \times 16 + 1 \times 8 + 0 \times 4 + 0 \times 2 + 1$, giving 11001.

31 (base 8), 100 (base 5), 11001 (base 2)

Example 7.2 What number does the base-5 numeral 2133 stand for?

SOLUTION

$2 \times 125 + 1 \times 25 + 3 \times 5 + 3 \times 1$

293

8 Common mistakes

- Writing 4 as IIII or 40 as XXXX instead of IV and XL, or reading IX as 11.
- Mixing up a base with a group size. A base needs **every** landmark number to be a power of the same number.
- Thinking the Egyptian system had place value. It had a base but used separate symbols.
- Using the digit n in base n. Base 5 uses only 0, 1, 2, 3, 4.
- Dropping a zero in the middle of a numeral: 3605 and 365 are different numbers.

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